

Axiomatic System

Axiom 1 (Infinite Adjunction). Beginning with a base set P_0 , whenever a set P_n has been constructed, there exists a fixed distinguished element

$$\alpha_{n+1} \notin P_n$$

which has not appeared at any preceding construction stage. Distinct stages use distinct elements. Hence the construction may continue indefinitely.

At this point the subscripts

$$0, 1, 2, \dots$$

serve only as construction-stage labels. Their arithmetic properties have not yet been assumed or defined.

Axiom 2 (Function Well-Definedness). A function $f : X \rightarrow Y$ maps every input $x \in X$ to exactly one output $f(x) \in Y$. We denote evaluation by

$$f(x) \rightarrow y.$$

Axiom 3 (Structural Induction). Let $\mathcal{P}$ be a family generated from a base object P_0 by repeated application of a function S . If a property holds for P_0 , and whenever it holds for a constructed set P_n it also holds for $S(P_n)$, then the property holds for every set generated from P_0 by repeated applications of S .

Definitions and Rules

1.1 Base Set

Let

$$P_0 = \emptyset.$$

The empty set serves as the base representation of the symbol

$$0.$$

Thus

$$E(P_0) = 0.$$

1.2 Canonical Finite Sets

Using Axiom 1, define the canonical sets recursively.

Whenever P_n has already been constructed, define

$$P_{n+1} = P_n \cup \{\alpha_{n+1}\},$$

where

$$\alpha_{n+1} \notin P_n.$$

Consequently,

$$\begin{aligned} P_0 &= \emptyset, \\ P_1 &= \{\alpha_1\}, \\ P_2 &= \{\alpha_1, \alpha_2\}, \\ P_3 &= \{\alpha_1, \alpha_2, \alpha_3\}, \end{aligned}$$

and so on.

Let $\mathcal{P}$ denote the smallest family containing P_0 and closed under this construction.

Thus

$$\mathcal{P} = \{P_0, P_1, P_2, P_3, \dots\}.$$

Because the element introduced at each stage is fixed, each construction stage has a unique output.

1.3 Output Symbols and the Cardinality Function

We now assign one output symbol to each construction stage.

Define

$$E : \mathcal{P} \rightarrow \{0, 1, 2, 3, \dots\}$$

recursively by

$$E(P_0) = 0,$$

and whenever P_{n+1} is the immediate successor of P_n , let $E(P_{n+1})$ be the next output symbol after $E(P_n)$.

Accordingly,

$$\begin{aligned} E(P_0) &= 0, \\ E(P_1) &= 1, \\ E(P_2) &= 2, \\ E(P_3) &= 3, \end{aligned}$$

and so on.

Hence, after these symbols have been generated,

$$E(P_n) = n.$$

The statement $E(P_n) = n$ therefore does not use a previously defined arithmetic operation. It associates the symbol generated at stage n with the canonical set generated at the same stage.

Since each P_n contains exactly the elements introduced during the first n nonzero construction stages, $E(P_n)$ is also the finite cardinality of P_n . Therefore, once the symbols have been constructed, we may write

$$E(P_n) = |P_n|.$$

The domain of E in this construction is $\mathcal{P}$, so E is never required to assign a finite natural-number output to an arbitrary infinite set.

1.4 Successor Function

Define

$$S : \mathcal{P} \rightarrow \mathcal{P}$$

by

$$S(P_n) = P_n \cup \{\alpha_{n+1}\} = P_{n+1}.$$

Because α_{n+1} is fixed at construction stage $n + 1$, S maps every canonical set to a unique canonical set and is therefore well-defined.

Applying S repeatedly gives

$$\begin{aligned} P_0 &= \emptyset, \\ S(P_0) &= P_1 = \{\alpha_1\}, \\ S(S(P_0)) &= P_2 = \{\alpha_1, \alpha_2\}, \\ S(S(S(P_0))) &= P_3 = \{\alpha_1, \alpha_2, \alpha_3\}. \end{aligned}$$

Evaluating with E ,

$$\begin{aligned} E(P_0) &= 0, \\ E(S(P_0)) &= 1, \\ E(S(S(P_0))) &= 2, \\ E(S(S(S(P_0)))) &= 3. \end{aligned}$$

For any already constructed natural-number symbol k , write

$$S^k(P)$$

for k repeated applications of S , with

$$S^0(P) = P.$$

1.5 Natural Numbers Definition

The natural numbers are defined to be exactly the output symbols produced by evaluating E on the canonical sets:

$$\mathbb{N} = \{E(P) \mid P \in \mathcal{P}\}.$$

Thus

$$\boxed{\mathbb{N} = \{0, 1, 2, 3, \dots\}}.$$

This definition does not require a statement of the form $k \in \mathbb{N}$ before $\mathbb{N}$ itself has been constructed.

Defining Addition

We define addition using the canonical sets, the successor function S , and the evaluation function E .

For $P_1, P_2 \in \mathcal{P}$, define

$$\boxed{A(P_1, P_2) = E(S^{E(P_2)}(P_1))}.$$

Thus $A(P_1, P_2)$ is a natural-number symbol, not a set.

We define the ordinary notation $+$ by

$$\boxed{E(P_1) + E(P_2) := A(P_1, P_2)}.$$

Therefore,

$$E(P_1) + E(P_2) = E(S^{E(P_2)}(P_1)).$$

Addition by Zero

Since

$$E(P_0) = 0,$$

we have

$$A(P_1, P_0) = E(S^{E(P_0)}(P_1)) = E(S^0(P_1)) = E(P_1).$$

Hence

$$\boxed{E(P_1) + 0 = E(P_1)}.$$

Also,

$$A(P_0, P_1) = E(S^{E(P_1)}(P_0)).$$

By construction, applying S to P_0 exactly $E(P_1)$ times produces P_1 . Therefore

$$A(P_0, P_1) = E(P_1),$$

so

$$\boxed{0 + E(P_1) = E(P_1)}.$$

Successor Compatibility

Suppose

$$A(P_1, P_2) = E(P_r).$$

Then

$$S^{E(P_2)}(P_1) = P_r.$$

Since $S(P_2)$ contains exactly one more constructed element than P_2 ,

$$S^{E(S(P_2))}(P_1) = S(S^{E(P_2)}(P_1)).$$

Therefore

$$A(P_1, S(P_2)) = E(S(S^{E(P_2)}(P_1))) = E(S(P_r)).$$

Likewise,

$$S^{E(P_1)}(S(P_2)) = S(S^{E(P_1)}(P_2)).$$

Thus taking one successor step in either input produces one successor step in the resulting canonical set.

Commutativity of Addition

Theorem 1 *For all $P_1, P_2 \in \mathcal{P}$,*

$$A(P_1, P_2) = A(P_2, P_1).$$

Consequently,

$$E(P_1) + E(P_2) = E(P_2) + E(P_1).$$

Fix P_1 . We use structural induction on P_2 .

For the base case,

$$P_2 = P_0.$$

Then

$$A(P_1, P_0) = E(P_1).$$

Also,

$$A(P_0, P_1) = E(S^{E(P_1)}(P_0)) = E(P_1).$$

Hence

$$A(P_1, P_0) = A(P_0, P_1).$$

Now assume

$$A(P_1, P_2) = A(P_2, P_1).$$

Let

$$P'_2 = S(P_2).$$

By successor compatibility,

$$A(P_1, P'_2)$$

is the output symbol immediately following

$$A(P_1, P_2).$$

Likewise,

$$A(P'_2, P_1)$$

is the output symbol immediately following

$$A(P_2, P_1).$$

By the induction hypothesis,

$$A(P_1, P_2) = A(P_2, P_1),$$

so their immediate successor symbols are equal. Therefore

$$A(P_1, P'_2) = A(P'_2, P_1).$$

By structural induction,

$$A(P_1, P_2) = A(P_2, P_1)$$

for every $P_2 \in \mathcal{P}$.

Therefore it is valid to write

$$\boxed{E(P_1) + E(P_2) = E(P_2) + E(P_1)}.$$

Example: $1 + 2$

Let

$$P_0 = \emptyset,$$

$$P_1 = S(P_0),$$

$$P_2 = S(S(P_0)).$$

Then

$$E(P_1) = 1, \quad E(P_2) = 2.$$

Evaluate

$$A(P_1, P_2) = E \left(S^{E(P_2)}(P_1) \right).$$

Since

$$E(P_2) = 2,$$

we obtain

$$A(P_1, P_2) = E(S^2(P_1)).$$

But

$$S^2(P_1) = S(S(P_1)) = P_3.$$

Thus

$$A(P_1, P_2) = E(P_3) = 3.$$

Therefore

$$1 + 2 = 3.$$

By the commutativity theorem,

$$A(P_2, P_1) = A(P_1, P_2) = 3,$$

so

$$2 + 1 = 3.$$

Hence

$$\boxed{1 + 2 = 2 + 1 = 3.}$$

Subtraction on Natural Numbers via Element Removal

We now define subtraction when the result remains nonnegative.

1. Element Removal Function

Define the removal function N on every nonempty canonical set by

$$\boxed{N(P_n) = P_{n-1} \quad \text{for } P_n \neq P_0.}$$

Equivalently, N removes the distinguished element introduced most recently in the canonical construction.

For example,

$$P_3 = \{\alpha_1, \alpha_2, \alpha_3\}.$$

Then

$$N(P_3) = \{\alpha_1, \alpha_2\} = P_2.$$

Similarly,

$$N(P_2) = P_1$$

and

$$N(P_1) = P_0.$$

In particular,

$$N(P_1) = P_0 = \emptyset,$$

and therefore

$$E(N(P_1)) = 0.$$

The rule specifies a unique element to remove, so N is well-defined.

2. Iterated Removal

For any natural-number symbol k , define

$$N^0(P) = P,$$

and, whenever the intermediate set is nonempty,

$$N^k(P) = N(N^{k-1}(P))$$

for nonzero k .

For example,

$$N^2(P_3) = N(N(P_3)) = N(P_2) = P_1.$$

3. Ordering of Natural Numbers

Before restricting subtraction, define the order relation $\leq$.

For canonical sets P_1, P_2 , define

$$\boxed{E(P_2) \leq E(P_1)}$$

to mean that P_1 can be obtained from P_2 by zero or more applications of S .

Equivalently,

$$E(P_2) \leq E(P_1)$$

when there exists a natural-number symbol k such that

$$S^k(P_2) = P_1.$$

Thus, for example,

$$1 \leq 3$$

because

$$S^2(P_1) = P_3.$$

4. Definition of Natural Subtraction

Let

$$D(P_1, P_2)$$

be defined whenever

$$E(P_2) \leq E(P_1).$$

Define

$$\boxed{D(P_1, P_2) = E(N^{E(P_2)}(P_1))}.$$

We then define

$$\boxed{E(P_1) - E(P_2) := D(P_1, P_2)}$$

whenever

$$E(P_2) \leq E(P_1).$$

The restriction guarantees that N is never required to remove an element from P_0 .

5. Example Evaluation

Let

$$P_3 = \{\alpha_1, \alpha_2, \alpha_3\}$$

and

$$P_1 = \{\alpha_1\}.$$

Then

$$E(P_3) = 3$$

and

$$E(P_1) = 1.$$

Evaluate

$$D(P_3, P_1) = E(N^{E(P_1)}(P_3)).$$

Since

$$E(P_1) = 1,$$

we have

$$D(P_3, P_1) = E(N(P_3)).$$

By the definition of N ,

$$N(P_3) = P_2.$$

Therefore

$$D(P_3, P_1) = E(P_2) = 2.$$

Thus

$$\boxed{3 - 1 = 2}.$$

Constructing Negative Numbers

Natural-number subtraction as defined above does not yet assign an output to expressions such as

$$1 - 2.$$

We now construct signed integers using ordered pairs of canonical sets.

1. Ordered-Pair Representation

Take an ordered pair

$$(P_x, P_y),$$

where

$$P_x, P_y \in \mathcal{P}.$$

The first component records a positive contribution, and the second component records a negative contribution.

At this stage, however, the symbol “negative” describes the role of the second component only. A negative-number symbol such as -1 will be introduced only after the corresponding ordered-pair object has been constructed.

2. Reduction Function

Define

$$R[(P_x, P_y)]$$

recursively by

$$R[(P_x, P_y)] = \begin{cases} R[(N(P_x), N(P_y))], & E(P_x) > 0 \text{ and } E(P_y) > 0, \\ (P_x, P_y), & E(P_x) = 0 \text{ or } E(P_y) = 0. \end{cases}$$

Here

$$E(P) > 0$$

means simply

$$P \neq P_0.$$

Every recursive step removes one element from both components.

Since P_x and P_y are finite canonical sets, after finitely many applications of N , at least one component must become P_0 .

Therefore R always terminates.

Most importantly,

$$R$$

always returns an ordered pair. Its output type does not change during the definition.

3. Reduced Signed Pairs

A pair

$$(P_x, P_y)$$

is called *reduced* if at least one component equals P_0 .

Thus every reduced pair has exactly one of the following forms:

$$(P_n, P_0),$$

$$(P_0, P_n),$$

or

$$(P_0, P_0).$$

The function R maps every ordered pair of canonical sets to exactly one reduced pair.

4. Signed Integer Notation

We now introduce notation for these already-constructed reduced pairs.

For a nonzero natural-number symbol n , define the notation

$$(P_n, P_0) \equiv n.$$

Define

$$(P_0, P_0) \equiv 0.$$

Finally, for a nonzero natural-number symbol n , introduce the new notation

$$\boxed{(P_0, P_n) \equiv -n.}$$

Thus the symbol

$$-n$$

is not used to construct the negative integer.

Instead, the ordered pair

$$(P_0, P_n)$$

is constructed first, and $-n$ is then assigned as notation for that object.

The signed integers are therefore the reduced pairs

$$\boxed{\mathbb{Z} = \{R[(P_x, P_y)] \mid P_x, P_y \in \mathcal{P}\}.$$

Using the notation above, these objects are written

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

5. Canonical Representation

Because R removes one element from each side simultaneously, every pair has a unique reduced representative.

For example,

$$R[(P_2, P_1)] = R[(N(P_2), N(P_1))] = R[(P_1, P_0)] = (P_1, P_0).$$

Thus

$$(P_2, P_1)$$

and

$$(P_1, P_0)$$

reduce to the same signed integer.

More generally, two ordered pairs

$$(P_a, P_b)$$

and

$$(P_c, P_d)$$

represent the same signed integer precisely when

$$\boxed{R[(P_a, P_b)] = R[(P_c, P_d)].}$$

Hence signed integers have unique reduced representatives even though unreduced intermediate pairs may differ.

6. Examples

Example 1: $1 - 2$.

Take

$$P_1 = \{\alpha_1\}$$

and

$$P_2 = \{\alpha_1, \alpha_2\}.$$

Evaluate

$$R[(P_1, P_2)].$$

Since both components are nonempty,

$$R[(P_1, P_2)] = R[(N(P_1), N(P_2))].$$

Using

$$N(P_1) = P_0$$

and

$$N(P_2) = P_1,$$

we obtain

$$R[(P_1, P_2)] = R[(P_0, P_1)].$$

Because the first component is now P_0 ,

$$R[(P_0, P_1)] = (P_0, P_1).$$

By the signed notation introduced above,

$$(P_0, P_1) \equiv -1.$$

Therefore

$$\boxed{1 - 2 = -1.}$$

Example 2: $2 - 1$.

Evaluate

$$R[(P_2, P_1)].$$

Since both components are nonempty,

$$R[(P_2, P_1)] = R[(N(P_2), N(P_1))].$$

Thus

$$R[(P_2, P_1)] = R[(P_1, P_0)] = (P_1, P_0).$$

By signed notation,

$$(P_1, P_0) \equiv 1.$$

Therefore

$$\boxed{2 - 1 = 1.}$$

Example 3: $2 - 2$.

Evaluate

$$R[(P_2, P_2)].$$

First,

$$R[(P_2, P_2)] = R[(N(P_2), N(P_2))] = R[(P_1, P_1)].$$

Both components remain nonempty, so

$$R[(P_1, P_1)] = R[(N(P_1), N(P_1))] = R[(P_0, P_0)].$$

The pair is now reduced, giving

$$R[(P_0, P_0)] = (P_0, P_0).$$

Since

$$(P_0, P_0) \equiv 0,$$

we obtain

$$\boxed{2 - 2 = 0.}$$

Addition and Subtraction of Signed Pairs

We now define arithmetic directly on reduced signed-pair representations.

Let

$$(P_a, P_b)$$

and

$$(P_c, P_d)$$

be signed-pair representations.

The positive contributions are represented by

$$P_a \quad \text{and} \quad P_c,$$

while the negative contributions are represented by

$$P_b \quad \text{and} \quad P_d.$$

Signed Addition

The combined positive count is

$$A(P_a, P_c).$$

The corresponding canonical set is

$$S^{A(P_a, P_c)}(P_0).$$

Likewise, the combined negative count is

$$A(P_b, P_d),$$

with corresponding canonical set

$$S^{A(P_b, P_d)}(P_0).$$

Therefore define

$$\text{Add}((P_a, P_b), (P_c, P_d)) = R \left[(S^{A(P_a, P_c)}(P_0), S^{A(P_b, P_d)}(P_0)) \right].$$

The result is a reduced signed pair.

Signed Additive Inverse

For a signed pair

$$(P_c, P_d),$$

define its additive inverse by exchanging its two components:

$$-(P_c, P_d) := (P_d, P_c).$$

This is valid because the component that formerly contributed positively now contributes negatively, and the component that formerly contributed negatively now contributes positively.

For example,

$$-(P_2, P_0) = (P_0, P_2),$$

so the additive inverse of 2 is represented by

$$(P_0, P_2) \equiv -2.$$

Similarly,

$$-(P_0, P_2) = (P_2, P_0),$$

so the additive inverse of -2 is 2.

Signed Subtraction

Subtraction is addition of the additive inverse.

Thus

$$(P_a, P_b) - (P_c, P_d)$$

is defined as

$$(P_a, P_b) + (P_d, P_c).$$

Therefore

$$\text{Subtract}((P_a, P_b), (P_c, P_d)) = R \left[(S^{A(P_a, P_d)}(P_0), S^{A(P_b, P_c)}(P_0)) \right].$$

1. Adding Same Signs: $(-2) + (-1)$

The inputs are

$$(P_0, P_2)$$

and

$$(P_0, P_1).$$

The new positive count is

$$A(P_0, P_0) = 0.$$

Therefore

$$P_{\text{pos}} = S^0(P_0) = P_0.$$

The new negative count is

$$A(P_2, P_1) = 3.$$

Therefore

$$P_{\text{neg}} = S^3(P_0) = P_3.$$

Hence

$$\text{Add}((P_0, P_2), (P_0, P_1)) = R[(P_0, P_3)].$$

Since the first component is already P_0 ,

$$R[(P_0, P_3)] = (P_0, P_3).$$

By signed notation,

$$(P_0, P_3) \equiv -3.$$

Therefore

$$\boxed{(-2) + (-1) = -3.}$$

2. Adding Opposite Signs: $(-2) + 1$

The inputs are

$$(P_0, P_2)$$

and

$$(P_1, P_0).$$

The new positive count is

$$A(P_0, P_1) = 1,$$

so

$$P_{\text{pos}} = P_1.$$

The new negative count is

$$A(P_2, P_0) = 2,$$

so

$$P_{\text{neg}} = P_2.$$

Thus

$$\text{Add}((P_0, P_2), (P_1, P_0)) = R[(P_1, P_2)].$$

Applying N once to both components gives

$$R[(P_1, P_2)] = R[(P_0, P_1)] = (P_0, P_1).$$

Since

$$(P_0, P_1) \equiv -1,$$

we obtain

$$\boxed{(-2) + 1 = -1.}$$

3. Subtracting Same Signs: $1 - 2$

The inputs are

$$(P_1, P_0)$$

and

$$(P_2, P_0).$$

Using the subtraction definition,

$$A(P_1, P_0) = 1,$$

so

$$P_{\text{pos}} = P_1.$$

Also,

$$A(P_0, P_2) = 2,$$

so

$$P_{\text{neg}} = P_2.$$

Thus

$$\text{Subtract}((P_1, P_0), (P_2, P_0)) = R[(P_1, P_2)].$$

Reduction gives

$$R[(P_1, P_2)] = R[(P_0, P_1)] = (P_0, P_1).$$

Since

$$(P_0, P_1) \equiv -1,$$

we obtain

$$\boxed{1 - 2 = -1.}$$

4. Subtracting Opposite Signs: $1 - (-2)$

The inputs are

$$(P_1, P_0)$$

and

$$(P_0, P_2).$$

The positive count is

$$A(P_1, P_2) = 3,$$

so

$$P_{\text{pos}} = P_3.$$

The negative count is

$$A(P_0, P_0) = 0,$$

so

$$P_{\text{neg}} = P_0.$$

Therefore

$$\text{Subtract } ((P_1, P_0), (P_0, P_2)) = R[(P_3, P_0)].$$

Since the second component is already P_0 ,

$$R[(P_3, P_0)] = (P_3, P_0).$$

By signed notation,

$$(P_3, P_0) \equiv 3.$$

Hence

$$\boxed{1 - (-2) = 3.}$$

Proof by Fully Expanded Substitution

We now expand the preceding signed arithmetic computations back to the base set P_0 using only

$$S(P), \quad E(P), \quad A(P_1, P_2), \quad N(P), \quad R[(P_x, P_y)].$$

1. Adding Same Signs: $(-2) + (-1) = -3$

The base representations are

$$\text{Pair 1} = (P_0, S(S(P_0)))$$

and

$$\text{Pair 2} = (P_0, S(P_0)).$$

The expanded addition formula is

$$\text{Add} = R \left[(S^{A(P_0, P_0)}(P_0), S^{A(S(S(P_0)), S(P_0))}(P_0)) \right].$$

First evaluate the positive component:

$$A(P_0, P_0) = E(S^{E(P_0)}(P_0)).$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(P_0, P_0) = E(S^0(P_0)) = E(P_0) = 0.$$

Thus

$$k_{\text{pos}} = 0.$$

Now evaluate the negative component:

$$A(S(S(P_0)), S(P_0)) = E(S^{E(S(P_0))}(S(S(P_0)))).$$

Since

$$E(S(P_0)) = 1,$$

we obtain

$$A(S(S(P_0)), S(P_0)) = E(S^1(S(S(P_0)))).$$

Therefore

$$A(S(S(P_0)), S(P_0)) = E(S(S(S(P_0)))) = 3.$$

Thus

$$k_{\text{neg}} = 3.$$

Construct the new sets:

$$P_x = S^0(P_0) = P_0,$$

and

$$P_y = S^3(P_0) = S(S(S(P_0))).$$

Therefore

$$\text{Add} = R[(P_0, S(S(S(P_0))))].$$

Since the first component is P_0 , reduction stops immediately:

$$R[(P_0, S(S(S(P_0))))] = (P_0, S(S(S(P_0)))).$$

But

$$S(S(S(P_0))) = P_3.$$

Hence the reduced pair is

$$(P_0, P_3).$$

By signed notation,

$$(P_0, P_3) \equiv -3.$$

Therefore

$$\boxed{(-2) + (-1) = -3.}$$

2. Adding Opposite Signs: $(-2) + 1 = -1$

The base representations are

$$\text{Pair 1} = (P_0, S(S(P_0)))$$

and

$$\text{Pair 2} = (S(P_0), P_0).$$

The expanded addition formula is

$$\text{Add} = R \left[(S^{A(P_0, S(P_0))}(P_0), S^{A(S(S(P_0)), P_0)}(P_0)) \right].$$

Evaluate the positive component:

$$A(P_0, S(P_0)) = E(S^{E(S(P_0))}(P_0)).$$

Since

$$E(S(P_0)) = 1,$$

we have

$$A(P_0, S(P_0)) = E(S^1(P_0)) = E(S(P_0)) = 1.$$

Thus

$$k_{\text{pos}} = 1.$$

Now evaluate the negative component:

$$A(S(S(P_0)), P_0) = E(S^{E(P_0)}(S(S(P_0)))) .$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(S(S(P_0)), P_0) = E(S^0(S(S(P_0)))) = E(S(S(P_0))) = 2.$$

Thus

$$k_{\text{neg}} = 2.$$

Construct the sets:

$$P_x = S^1(P_0) = S(P_0),$$

and

$$P_y = S^2(P_0) = S(S(P_0)).$$

Therefore

$$\text{Add} = R[(S(P_0), S(S(P_0)))].$$

Because both sets are nonempty,

$$R[(S(P_0), S(S(P_0)))] = R[(N(S(P_0)), N(S(S(P_0))))].$$

By the definition of N ,

$$N(S(P_0)) = P_0$$

and

$$N(S(S(P_0))) = S(P_0).$$

Therefore

$$R[(S(P_0), S(S(P_0)))] = R[(P_0, S(P_0))].$$

The first component is now P_0 , so

$$R[(P_0, S(P_0))] = (P_0, S(P_0)).$$

Since

$$S(P_0) = P_1,$$

the reduced pair is

$$(P_0, P_1).$$

By signed notation,

$$(P_0, P_1) \equiv -1.$$

Therefore

$$\boxed{(-2) + 1 = -1.}$$

3. Subtracting Same Signs: $1 - 2 = -1$

The base representations are

$$\text{Pair 1} = (S(P_0), P_0)$$

and

$$\text{Pair 2} = (S(S(P_0)), P_0).$$

The expanded subtraction formula is

$$\text{Subtract} = R \left[(S^{A(S(P_0), P_0)}(P_0), S^{A(P_0, S(S(P_0)))}(P_0)) \right].$$

Evaluate the positive component:

$$A(S(P_0), P_0) = E(S^{E(P_0)}(S(P_0))) .$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(S(P_0), P_0) = E(S^0(S(P_0))) = E(S(P_0)) = 1.$$

Thus

$$k_{\text{pos}} = 1.$$

Evaluate the negative component:

$$A(P_0, S(S(P_0))) = E(S^{E(S(P_0))}(P_0)) .$$

Since

$$E(S(S(P_0))) = 2,$$

we obtain

$$A(P_0, S(S(P_0))) = E(S^2(P_0)) = E(S(S(P_0))) = 2.$$

Thus

$$k_{\text{neg}} = 2.$$

Construct the sets:

$$P_x = S^1(P_0) = S(P_0),$$

and

$$P_y = S^2(P_0) = S(S(P_0)).$$

Therefore

$$\text{Subtract} = R[(S(P_0), S(S(P_0)))].$$

Reduction gives

$$R[(S(P_0), S(S(P_0)))] = R[(P_0, S(P_0))] = (P_0, S(P_0)).$$

Since

$$S(P_0) = P_1,$$

we obtain

$$(P_0, S(P_0)) = (P_0, P_1).$$

By signed notation,

$$(P_0, P_1) \equiv -1.$$

Therefore

$$\boxed{1 - 2 = -1.}$$

4. Subtracting Opposite Signs: $1 - (-2) = 3$

The base representations are

$$\text{Pair 1} = (S(P_0), P_0)$$

and

$$\text{Pair 2} = (P_0, S(S(P_0))).$$

The expanded subtraction formula is

$$\text{Subtract} = R \left[(S^{A(S(P_0), S(S(P_0)))}(P_0), S^{A(P_0, P_0)}(P_0)) \right].$$

Evaluate the positive component:

$$A(S(P_0), S(S(P_0))) = E(S^{E(S(S(P_0)))}(S(P_0))).$$

Since

$$E(S(S(P_0))) = 2,$$

we obtain

$$A(S(P_0), S(S(P_0))) = E(S^2(S(P_0))).$$

Therefore

$$A(S(P_0), S(S(P_0))) = E(S(S(S(P_0)))) = 3.$$

Thus

$$k_{\text{pos}} = 3.$$

Evaluate the negative component:

$$A(P_0, P_0) = E(S^{E(P_0)}(P_0)).$$

Since

$$E(P_0) = 0,$$

we obtain

$$A(P_0, P_0) = E(P_0) = 0.$$

Thus

$$k_{\text{neg}} = 0.$$

Construct the sets:

$$P_x = S^3(P_0) = S(S(S(P_0))),$$

and

$$P_y = S^0(P_0) = P_0.$$

Therefore

$$\text{Subtract} = R[(S(S(S(P_0))), P_0)].$$

Since the second component is already P_0 ,

$$R[(S(S(S(P_0))), P_0)] = (S(S(S(P_0))), P_0).$$

Since

$$S(S(S(P_0))) = P_3,$$

the reduced pair is

$$(P_3, P_0).$$

By signed notation,

$$(P_3, P_0) \equiv 3.$$

Therefore

$$\boxed{1 - (-2) = 3.}$$

Conclusion

Beginning with

$$P_0 = \emptyset,$$

the canonical sets are generated by the well-defined successor function

$$S(P_n) = P_{n+1}.$$

The evaluation function E associates each canonical construction stage with exactly one natural-number symbol:

$$E(P_n) = n.$$

This produces

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}.$$

Addition is then defined by

$$\boxed{A(P_1, P_2) = E\left(S^{E(P_2)}(P_1)\right),}$$

with

$$E(P_1) + E(P_2) := A(P_1, P_2).$$

Commutativity follows by structural induction:

$$\boxed{E(P_1) + E(P_2) = E(P_2) + E(P_1).}$$

For nonnegative results, subtraction is defined by the deterministic removal function

$$N(P_n) = P_{n-1}$$

and

$$D(P_1, P_2) = E(N^{E(P_2)}(P_1)).$$

Signed integers are then constructed as reduced ordered pairs of canonical sets. The reduction function

$$R[(P_x, P_y)] = \begin{cases} R[(N(P_x), N(P_y))], & E(P_x) > 0 \text{ and } E(P_y) > 0, \\ (P_x, P_y), & E(P_x) = 0 \text{ or } E(P_y) = 0 \end{cases}$$

always returns a reduced ordered pair.

Only after these ordered-pair objects have been constructed is signed notation introduced:

$$(P_n, P_0) \equiv n,$$

$$(P_0, P_0) \equiv 0,$$

and

$$(P_0, P_n) \equiv -n.$$

Signed addition is defined by

$$\text{Add}((P_a, P_b), (P_c, P_d)) = R[(S^{A(P_a, P_c)}(P_0), S^{A(P_b, P_d)}(P_0))],$$

while signed subtraction is defined by

$$\text{Subtract}((P_a, P_b), (P_c, P_d)) = R[(S^{A(P_a, P_d)}(P_0), S^{A(P_b, P_c)}(P_0))].$$

The explicit evaluations establish

$$1 + 2 = 3,$$

$$3 - 1 = 2,$$

$$1 - 2 = -1,$$

$$2 - 1 = 1,$$

$$2 - 2 = 0,$$

$$(-2) + (-1) = -3,$$

$$(-2) + 1 = -1,$$

and

$$1 - (-2) = 3.$$

Thus the natural-number and signed-integer operations used above are obtained from the base set P_0 , deterministic adjunction and removal, evaluation of canonical finite sets, and reduction of ordered pairs.